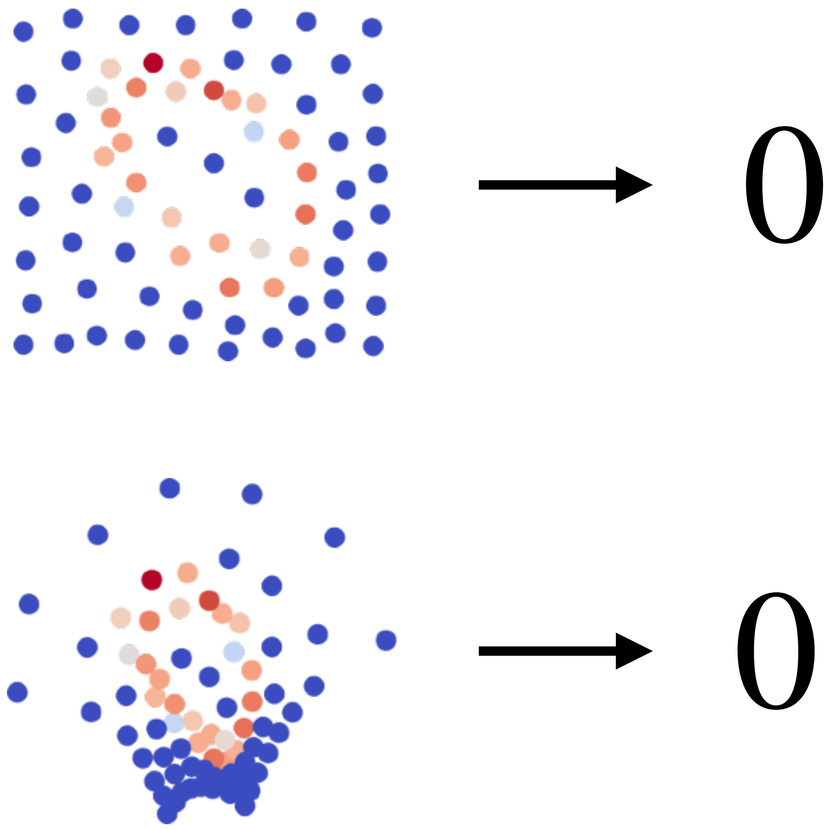
Conformally equivariant neural networks

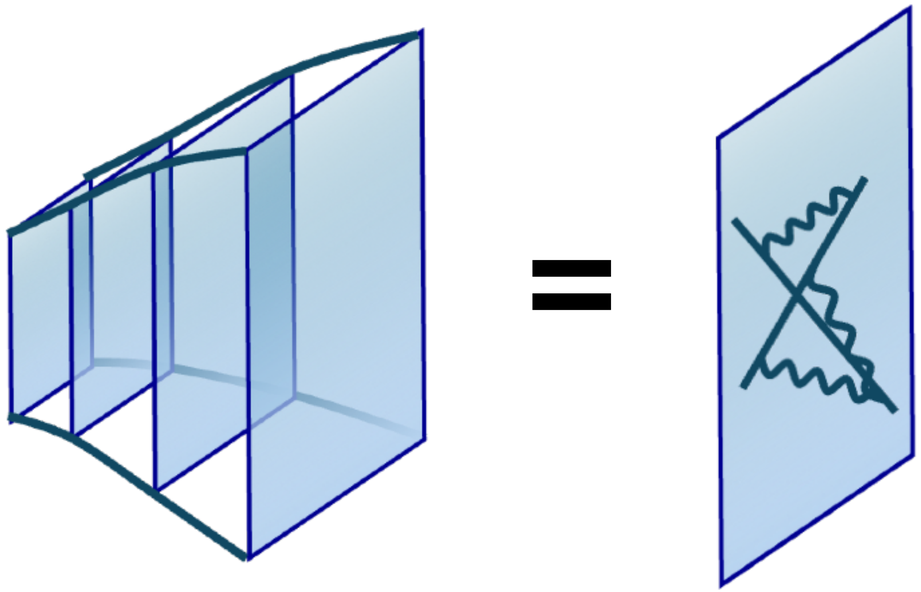
- We want to build a neural network that takes in a cloud of points and is **invariant** under conformal transformations: so we get exactly the same answer if we conformally transform each point.

- What is an **efficient way to process the point cloud** in a way that transforms nicely under conformal symmetry?

- Standard methods are difficult (conformal group is **non-compact** and **high-dimensional**)



Random interlude: AdS/CFT correspondence



$$ds^2 = \frac{1}{z^2}(dz^2 + dx^2 + dy^2)$$

- The AdS_{d+1} / CFT_d correspondence from string theory tells us that sometimes theories of particles in d-dimensions are exactly dual to theories of quantum gravity living in Anti de Sitter space AdS_{d+1}.
- This **holographic duality** is a deep and profound fact about quantum gravitational dynamics.
- The most basic kinematics of this is that the **conformal group** in d dimensions is the same as the **isometry group** of AdS_{d+1}.
- More precisely: AdS_{d+1} has a $\mathbb{R}^d$ boundary, and its isometry group acts as the conformal group on points at the boundary.